FIZ 101 · Physics I: Mechanics · Fall 2026 · Week 6

Potential energy and energy conservation

Lecture notes · Mehmet Özkan, Istanbul Technical University

8 Potential energy and energy conservation

In the previous chapter we saw why conserved quantities are so useful in Newtonian mechanics (Section 7.1), and we met kinetic energy and the work–energy theorem (Sections 7.2 and 7.3). In this chapter we come to the central idea: the *conservation of energy*. On the way we need a few concepts that are important in their own right and that we will use again and again: potential energy (Section 8.1), conservative forces (Section 8.2), energy diagrams and equilibrium (Section 8.3), the relation between force and potential energy in three dimensions (Section 8.4), and nonconservative forces such as friction (Section 8.5).

Recall the work–energy theorem (7.7): when a body moves from a point A to a point B,

$$W_{\text{net}} = K_B - K_A, \qquad K = \tfrac{1}{2}mv^2,$$

where W_{net} is the *total* work done on the body by all the forces that act on it (the work of the net force). Throughout this chapter we work in an inertial frame, treat bodies as particles, and take $g = 9.8\,\text{m/s}^2$; unless we say otherwise, we neglect friction and air resistance.

8.1 Gravitational potential energy

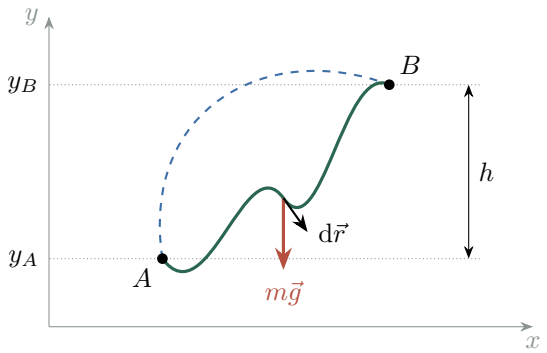


Figure 8.1. A body moves from A to B, a height $h = y_B - y_A$ higher. The work done on it by gravity is $-mgh$ along either path, the solid one or the dashed one: only the vertical part of each displacement $d\vec{r}$ counts.

Consider a body of mass m that moves from a point A to a point B whose height above A is $h = y_B - y_A$ (h is negative if B is below A), as in Figure 8.1, under the action of gravity and possibly of other forces. Near the surface of the Earth gravity is a constant force, $\vec{F} = -mg\,\hat{\jmath}$ (we take the $+y$ axis upward). By (7.21) and (7.24), its work along *any*

path from A to B is

$$W_{AB} = \int_A^B \vec{F} \cdot \mathrm{d}\vec{r} = \int_{y_A}^{y_B} (-mg)\,\mathrm{d}y = -mg(y_B - y_A),$$

because the horizontal parts of the displacement contribute nothing: gravity is perpendicular to them. So the work done by gravity is

$$W_{AB} = F_y(y_B - y_A) = -mgh. \tag{8.1}$$

It is negative if B is higher than A ($h > 0$: the body rises, against the force), positive if B is lower ($h < 0$), and zero if A and B are at the same height, whatever the path between them.

Now suppose that gravity is the only force that does work on the body. Other forces may act, as long as they do no work: the normal force of a fixed, frictionless track, for example, is always perpendicular to the velocity. Then the net work is the work of gravity, and by the work–energy theorem (7.7) it equals the change in the kinetic energy of the body:

$$-mgh = -mg(y_B - y_A) = K_B - K_A. \tag{8.2}$$

If we collect the quantities that belong to A on one side and those that belong to B on the other, this becomes

$$mgy_A + K_A = mgy_B + K_B. \tag{8.3}$$

The quantity mgy is called the **gravitational potential energy**:

$$U_{\text{grav}} = mgy. \tag{8.4}$$

Equation (8.3) says that the sum of the kinetic and potential energies of the body at A is equal to the same sum at B. We call this sum the **mechanical energy**, $E = K + U$. As the body moves, potential energy can turn into kinetic energy, and kinetic energy into potential energy, but their sum, the mechanical energy, stays constant. In terms of U, the work done by gravity is

$$W_{\text{grav}} = U_A - U_B = -\Delta U.$$

When the body rises, gravity does negative work and U increases. The energy is not lost: it is stored, and gravity gives it back as kinetic energy when the body comes down. This is the meaning of the word "potential".

Two remarks. First, the potential energy is not really a property of the body alone. It depends on the separation between the body and the Earth, and it is the energy of their gravitational interaction, so it belongs to the *system* of the body and the Earth. The Earth is so massive that its own motion is negligible, so all the change in kinetic energy is the body's, and we usually speak of "the potential energy of the body" for short. Second, $U = mgy$ holds only near the surface of the Earth, where g is constant. When the height is not small compared with the radius of the Earth (6370 km), we need the general expression that follows from Newton's law of gravitation: already at 400 km, the height of the International Space Station, g is more than 10% smaller than at the surface.

Looking at the definition (8.4), we see that the potential energy is zero at $y = 0$. But where is this point $y = 0$? Can it be the floor of our classroom, the center of the Earth,

or the roof of our house? The law of conservation of mechanical energy (8.3) tells us that what matters is not the potential energy itself, but the difference in potential energy, $\Delta U = U_B - U_A = mg(y_B - y_A)$. This depends only on the difference in height between A and B, not on where $y = 0$ is. Indeed, if we measure heights from another level, $y' = y - y_0$, every potential energy changes by the same constant, $-mgy_0$, and this constant appears on both sides of (8.3) and cancels. Moving the level $y = 0$ changes U, and with it the mechanical energy, by the same constant at every point, but it changes no difference ΔU, and so no speed, kinetic energy or force. So the choice of $y = 0$ cannot change the result of a problem, and we are free to put it wherever it makes the problem simplest, as long as we keep the same choice throughout the problem. With a given choice, U may well be negative (for a body below the level $y = 0$); that is no problem, since only differences of U have a physical meaning.

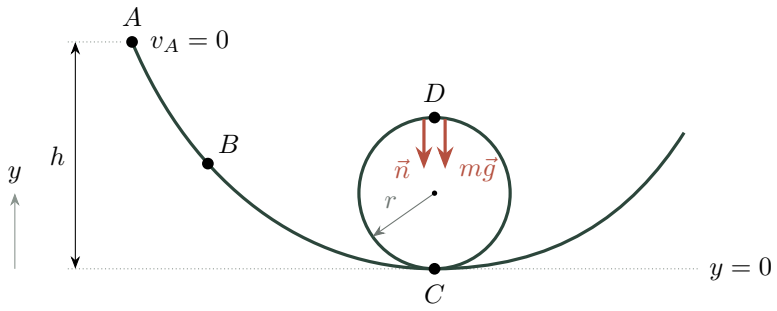


Figure 8.2. A roller coaster with a circular loop of radius r. The car starts from rest at A, a height h above the lowest point C; D is the top of the loop. At D both forces on the car, its weight $m\vec{g}$ and the normal force $\vec{n}$ of the track, point down, toward the center of the loop.

To see what the conservation of mechanical energy can do, consider the roller coaster in Figure 8.2, with a loop of radius r. We release a car at A without giving it any initial speed ($v_A = 0$). It starts to slide down the track and picks up speed, turning potential energy into kinetic energy. We assume that the track is frictionless and that there is no air resistance. The track exerts a normal force on the car, but this force is always perpendicular to the velocity and does no work, so gravity is the only force that does work, and the mechanical energy is conserved:

$$K_A + U_A = K_B + U_B = K_C + U_C = K_D + U_D. \tag{8.5}$$

We choose $y = 0$ at the lowest point C. The car is at rest at A, which is a height h above $y = 0$, so its mechanical energy is

$$mgh = K_B + U_B = K_C + U_C = K_D + U_D. \tag{8.6}$$

At any point of the track at a height y above C, the speed v of the car is therefore given by

$$mgh = mgy + \frac{1}{2}mv^2, \qquad \Rightarrow \qquad v^2 = 2g(h - y). \tag{8.7}$$

The speed depends only on how far the car has come down from A, not on the shape of the track, and it is the same as for a body that falls freely through the same height. (The shape of the track does decide the *direction* of the velocity and how long the trip takes, and energy alone cannot tell us these.) Since v^2 cannot be negative, the car can never rise

above the height h from which it started. The car is fastest at the lowest point C, where U is smallest and so $K = E - U$ is largest; with our choice $y = 0$ at C, $U_C = 0$ and all the mechanical energy mgh is kinetic there: $v_C = \sqrt{2gh}$.

Equation (8.7) also gives the speed at D, provided the car gets there while staying on the track. After C the car moves on a circle, so it has a radial acceleration $a_{\text{rad}} = v^2/r$ toward the center of the loop; its speed changes along the loop, so this is nonuniform circular motion, and v is the speed at the point we consider. At the top, D, the car is upside down on the inside of the track, and both forces on it point down, toward the center: its weight mg and the normal force n of the track (Figure 8.2). Newton's second law in the radial direction gives $mv_D^2/r = n + mg$, so $n = m(v_D^2/r - g)$, just as for the tension of the string in (6.19). The track can push on the car but it cannot pull it, so $n \geq 0$ (real roller-coaster cars have wheels on both sides of the rail, so that the track can also pull them, but here we consider a car that simply runs on the inside of the track), and the car stays on the track at the top only if

$$a_{\text{rad}} = \frac{v_D^2}{r} \geq g. \tag{8.8}$$

The top of the loop is at $y = 2r$, so by (8.7) the speed of the car at D is given by

$$v_D^2 = 2g(h - 2r). \tag{8.9}$$

Putting this into (8.8), $2g(h - 2r)/r \geq g$, and so

$$2h - 4r \geq r, \qquad \Rightarrow \qquad h \geq \frac{5}{2}r. \tag{8.10}$$

Why is it enough to look at the top? At a point of the loop at an angle φ from the bottom, the car is at the height $y = r(1 - \cos\varphi)$, and the radial component of the weight, $mg\cos\varphi$, points away from the center. The second law in the radial direction, $n - mg\cos\varphi = mv^2/r$, together with (8.7), gives

$$n = \frac{mg}{r}\,(2h - 2r + 3r\cos\varphi),$$

which is smallest at the top, $\varphi = 180°$, where it is $n_D = mg(2h - 5r)/r$. So if the car stays on the track at D, it stays on it everywhere. If h is smaller than $\frac{5}{2}r$ but larger than r, n drops to zero somewhere in the upper half of the loop, before the car reaches D. From there on, gravity alone pulls the car toward the center more strongly than motion on the circle requires, so the car leaves the track and falls, following a projectile path. (If $h \leq r$, the car never rises above the center of the loop: it stops on the track and slides back down.)

The car must start at least half a radius above the top of the loop: this extra height gives it the speed $\sqrt{gr}$ that it needs at the top. Notice that the mass has cancelled. A real roller coaster has friction, so it must start higher still. Note also how hard the track must push at the bottom: for $h = \frac{5}{2}r$, $v_C^2 = 2gh = 5gr$, and the second law at C, $n_C - mg = mv_C^2/r$, gives $n_C = 6mg$. The riders would feel six times as heavy as usual, which is far too much; this is one reason why real loops are not circles but are tighter at the top than at the bottom.[1]

[1] We have treated the car as a particle that slides. A solid ball that *rolls* along the same track also has kinetic energy of rotation, so less of its energy is available for its forward motion; it needs $h \geq 2.7\,r$. We will see why when we study the rotation of rigid bodies.

Example 8.1: The vertical circle revisited

In Section 6.1 a body of mass m on a string of length R moved in a vertical circle, and we promised to show that the tension at the bottom point P is larger than the tension at the top point S by $T_P - T_S = 6mg$, whatever the speed. We assume that the other end of the string is held fixed at the center of the circle (a hand that moves can do work on the body through the string) and that air resistance is negligible. Then the tension is always along the string, perpendicular to the velocity, so it does no work, and the mechanical energy of the body is conserved. The top is a height $2R$ above the bottom, so

$$\tfrac{1}{2}mv_P^2 = \tfrac{1}{2}mv_S^2 + mg(2R), \qquad \Rightarrow \qquad v_P^2 = v_S^2 + 4gR.$$

From (6.17) and (6.19), with $a_{\text{rad}} = v^2/R$, the tensions are $T_P = m(v_P^2/R + g)$ and $T_S = m(v_S^2/R - g)$, so

$$T_P - T_S = \frac{m(v_P^2 - v_S^2)}{R} + 2mg = 4mg + 2mg = 6mg.$$

This holds as long as the string stays taut. For the roller coaster the normal force plays the role of the tension: with $h = \frac{5}{2}r$, $n_D = 0$ and so $n_C = 6mg$, as we found above.

8.2 Conservative forces and the conservation of mechanical energy

When we studied the gravitational potential energy, we saw that the work done by gravity when a body moves from a point A to a point B does not depend on the path it follows. When the work done by a force is independent of the path and depends only on the starting and ending points of the motion, like the work of gravity, we call that force a **conservative force**. An equivalent statement is that the work done by a conservative force around any *closed* path, one that returns to its starting point, is zero: what the force takes from the body on one part of the path, it gives back on another. (To see that the two statements are equivalent, go from A to B along one path and come back along another. On the way back the work is minus the work along the second path from A to B, so the total is zero exactly when the two paths give the same work.)

Only for a conservative force can we define a potential energy. Choose a reference point O and give U any value U_O there. For every point P, define $U_P = U_O - W_{O\to P}$, where $W_{O\to P}$ is the work done by the force along any path from O to P. This makes sense only because the work is the same along every path. Going from A to B by way of O, and using the fact that the work from A to O is minus the work from O to A, $W_{AB} = W_{A\to O} + W_{O\to B} = (U_A - U_O) + (U_O - U_B)$, so

$$W_{AB} = U_A - U_B = -\Delta U.$$

The choice of O and U_O is the free choice of the zero of potential energy that we met for gravity. For a force whose work depends on the path, no function of position can do this.

To see why conservative forces are called by this name, let us go back to equation (8.1): as the body moves, the work done on it by gravity, $-mgh$, is minus the change in its

potential energy, $W = -\Delta U$. In one dimension, the work done by a force F during a small displacement $\mathrm{d}x$ is $\mathrm{d}W = F\,\mathrm{d}x$, and for a conservative force it is also $\mathrm{d}W = -\mathrm{d}U$. So there is a relation between the force and the potential energy:

$$F = -\frac{\mathrm{d}U}{\mathrm{d}x}. \tag{8.11}$$

Conversely, we find the potential energy from the force by integrating:

$$U(x) = U(x_{\mathrm{ref}}) - \int_{x_{\mathrm{ref}}}^{x} F(x')\,\mathrm{d}x',$$

where the reference point x_{ref} and the value $U(x_{\mathrm{ref}})$ there are our free choice of the zero. The minus sign in (8.11) says that a conservative force always points toward lower potential energy, "downhill". Now consider the mechanical energy of the body,

$$E = \frac{1}{2}mv^2 + U, \tag{8.12}$$

and suppose that F is the only force on it. We take the time derivative of E, using the chain rule, $\mathrm{d}U/\mathrm{d}t = (\mathrm{d}U/\mathrm{d}x)(\mathrm{d}x/\mathrm{d}t) = v\,\mathrm{d}U/\mathrm{d}x$, where $v = \mathrm{d}x/\mathrm{d}t$ is the velocity along x, positive or negative:

$$\frac{\mathrm{d}E}{\mathrm{d}t} = mv\frac{\mathrm{d}v}{\mathrm{d}t} + \frac{\mathrm{d}U}{\mathrm{d}t} = v\Big(m\frac{\mathrm{d}v}{\mathrm{d}t} + \frac{\mathrm{d}U}{\mathrm{d}x}\Big) = 0. \tag{8.13}$$

The bracket vanishes because, by Newton's second law and (8.11), $m\,\mathrm{d}v/\mathrm{d}t = F = -\mathrm{d}U/\mathrm{d}x$. So conservative forces *conserve* the mechanical energy, and this is where their name comes from.

Notice where we used the fact that U depends only on the position x. If U also depended on the time, say because someone kept moving the support of a spring, $\mathrm{d}U/\mathrm{d}t$ would contain an extra term, the rate at which U changes at a fixed point, and E would no longer be constant. A force law that does not change in time is just the homogeneity of time of Section 5.2.1: an experiment done tomorrow gives the same result as one done today. As we said in Section 7.1, this symmetry is the reason why energy is conserved; here we have seen how it works.

So a force is conservative if there is a function $U(x)$ that gives it through (8.11). In one dimension, every force that depends only on the position x of the body passes this test, since we can always find U by integrating. Forces that depend on the velocity, such as friction and air resistance, whose direction is always opposite to the motion, do not (Section 8.5). As we saw in the previous section, gravity is a conservative force: from $U_{\mathrm{grav}} = mgy$, (8.11) gives $F_y = -\mathrm{d}(mgy)/\mathrm{d}y = -mg$, the weight, pointing down, toward lower U.

As another example, consider a harmonic oscillator, a body attached to an ideal spring. Its potential energy, the **elastic potential energy**, is

$$U_{\mathrm{el}} = \frac{1}{2}kx^2, \tag{8.14}$$

where k is the spring constant and x is the stretch of the spring, measured from its relaxed length ($x < 0$ for a compressed spring; U_{el} is positive either way). From (8.11) the spring force is then $F_{\text{el}} = -\mathrm{d}U_{\text{el}}/\mathrm{d}x = -kx$, which is exactly Hooke's law (7.26): the spring force is conservative. (We can also go the other way: starting from Hooke's law and choosing $U = 0$ at $x_{\text{ref}} = 0$, the integral above gives $U_{\text{el}} = -\int_0^x (-kx')\,\mathrm{d}x' = \frac{1}{2}kx^2$.) This agrees with what we found in Chapter 7. When we stretch a spring from $x = 0$ to x, the spring does the work $-\frac{1}{2}kx^2$ on the body (7.27), which is $-\Delta U_{\text{el}}$, while we do the work $+\frac{1}{2}kx^2$ on it (7.28); this work is stored in the spring as elastic potential energy, and the spring gives it back when it is released. Equation (7.30) for a body released from the stretch $x_{\max}$ is just the conservation of $E = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$: $\frac{1}{2}kx_{\max}^2 = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$.

When several conservative forces do work on a body, each has its own potential energy, and U is their sum. This gives the **law of conservation of mechanical energy**:

If only conservative forces do work on a body, its mechanical energy is conserved:

$$K_A + U_A = K_B + U_B, \qquad U = U_{\text{grav}} + U_{\text{el}} + \cdots .$$

Forces that do no work, such as the normal force of a fixed surface, do not change E.

Example 8.2: A spring gun

The spring of a toy gun has $k = 600\,\text{N/m}$. We compress it by $0.10\,\text{m}$, put a ball of mass $0.020\,\text{kg}$ on it and fire the ball straight up. How high does it go?

Two conservative forces do work on the ball, the spring force and gravity, so $E = K + U_{\text{el}} + U_{\text{grav}}$ is conserved. We put $y = 0$ at the starting point. At the start the ball is at rest and all the energy is elastic, $\frac{1}{2}kx^2 = \frac{1}{2}(600\,\text{N/m})(0.10\,\text{m})^2 = 3.0\,\text{J}$. At the highest point the ball is again at rest, the spring is relaxed and all the energy is gravitational, mgH. So

$$H = \frac{kx^2}{2mg} = \frac{3.0\,\text{J}}{(0.020\,\text{kg})(9.8\,\text{m/s}^2)} = 15\,\text{m}.$$

We did not need to know how the force of the spring changes during the launch, only the energy it stores. The ball leaves the spring when the spring reaches its relaxed length, $0.10\,\text{m}$ above the start; there $\frac{1}{2}mv^2 = 3.0\,\text{J} - mg(0.10\,\text{m}) = 2.98\,\text{J}$, so $v = 17\,\text{m/s}$.

One of the great advantages of conservative forces is that, instead of Newton's equations, which are *second-order* differential equations (they contain $\mathrm{d}^2x/\mathrm{d}t^2$), they give us a *first-order* differential equation, one that contains only $\mathrm{d}x/\mathrm{d}t$. If we want to study, in general, a system that moves in one dimension under a conservative force, we use (8.12) and find

$$E = \frac{1}{2}m\Big(\frac{\mathrm{d}x}{\mathrm{d}t}\Big)^2 + U(x), \qquad \Rightarrow \qquad \frac{\mathrm{d}x}{\mathrm{d}t} = \pm\sqrt{\frac{2}{m}(E - U(x))}, \tag{8.15}$$

where we take the $+$ sign while the body moves in the $+x$ direction and the $-$ sign while it moves in the $-x$ direction. As we found above, the mechanical energy E does not depend on time, and the potential energy is a function of position only. So we can separate the

variables, putting everything that contains x on one side and $\mathrm{d}t$ on the other, and integrate from the moment t_0, when the body is at x_0, to the moment t, when it is at x:

$$t - t_0 = \pm \int_{x_0}^{x} \frac{\mathrm{d}x'}{\sqrt{\frac{2}{m}(E - U(x'))}}. \tag{8.16}$$

If we can do this integral, we know t as a function of x; turning this around gives $x(t)$, and the problem is solved. Notice that the constants E and x_0, together with the choice of sign, play the role of the two initial conditions that Newton's second law needs: x_0 is the initial position, E gives the initial speed through $\frac{1}{2}mv_0^2 = E - U(x_0)$, and the sign gives the direction of the initial velocity. The sign in (8.16) changes at the points where $E = U(x)$, where the speed is zero and the body turns around; we come back to these *turning points* in the next section. (This trick works only in one dimension: in three dimensions, conservation of energy is one equation for three unknown coordinates.)

Example 8.3: A body on a spring

For a body on a spring, $U = \frac{1}{2}kx^2$. If the body oscillates between $x = -A$ and $x = +A$, it is at rest at these turning points, so $E = \frac{1}{2}kA^2$ and $\frac{2}{m}(E-U) = \frac{k}{m}(A^2-x^2)$. Suppose that the body passes through $x_0 = 0$ at $t_0 = 0$, moving in the $+x$ direction. With $\int \mathrm{d}x/\sqrt{A^2 - x^2} = \arcsin(x/A)$, (8.16) gives

$$t = \sqrt{\frac{m}{k}} \int_0^x \frac{\mathrm{d}x'}{\sqrt{A^2 - x'^2}} = \sqrt{\frac{m}{k}} \arcsin\frac{x}{A}, \qquad \Rightarrow \qquad x(t) = A\sin\omega t, \quad \omega = \sqrt{\frac{k}{m}}.$$

We derived this for the first quarter of a period, while the body moves from 0 to A. After the turning point at $x = A$ we must take the minus sign in (8.16), and the same steps show that $x = A\sin\omega t$ continues to hold, through every later turning point. So the body oscillates with period $2\pi/\omega = 2\pi\sqrt{m/k}$, independent of the amplitude A: this is *simple harmonic motion*. We have solved Newton's second law, $m\,\mathrm{d}^2x/\mathrm{d}t^2 = -kx$, without ever writing it down.

8.3 Energy diagrams

When a body moves along a line and only a conservative force does work on it, so that its mechanical energy is constant, the graph of its potential energy $U(x)$ shows the main features of the motion without our solving any equation. To see this, consider a potential energy of the form

$$U(x) = m(x^3 - 3x). \tag{8.17}$$

(This is a made-up example, chosen to keep the numbers simple: we measure x in meters and U in joules, and we do not write the units of the two coefficients, $\mathrm{m}^{-1}\,\mathrm{s}^{-2}$ and $\mathrm{m}\,\mathrm{s}^{-2}$. Since U is proportional to the mass, the acceleration $a_x = F/m = -(1/m)\,\mathrm{d}U/\mathrm{d}x = 3-3x^2$ does not depend on m, just as in free fall.) Its graph is shown in Figure 8.3. It has a local maximum at $x = -1$, where $U = 2m$, and a local minimum at $x = +1$, where $U = -2m$; it also takes the value $2m$ at $x = 2$.

Of course, with this potential energy we could do the integral (8.16); the result is an elliptic integral, a function that is not among the ones we know from calculus. But we can say a

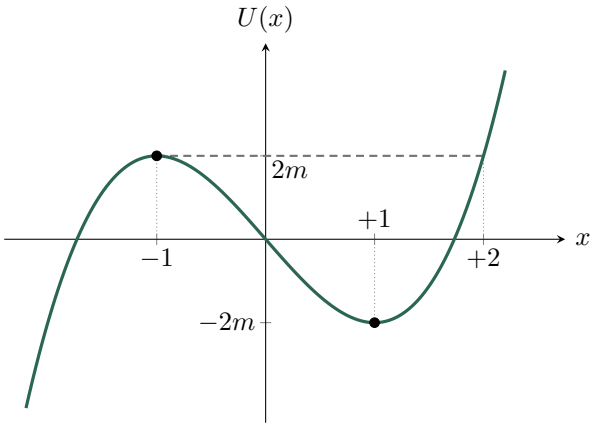


Figure 8.3. The potential energy $U(x) = m(x^3 - 3x)$. It has a local maximum $U = 2m$ at $x = -1$ and a local minimum $U = -2m$ at $x = +1$; the dashed line shows that $U(2)$ is equal to $U(-1)$.

lot about the motion without doing the integral. The key is that the kinetic energy can never be negative:

$$K = \frac{1}{2}mv^2 = E - U(x) \geq 0.$$

So if we draw a horizontal line at the height E on the graph of $U(x)$, the body can only be where the curve is *below* the line, and the vertical distance from the curve up to the line is its kinetic energy at that point. Where the curve meets the line, $U = E$, the speed is zero: these are the **turning points**, where the body stops for an instant and turns around. Meanwhile the force $F = -\mathrm{d}U/\mathrm{d}x$ is minus the slope of the curve: it is strongest where the curve is steepest, and it always points downhill.

Now suppose that the body starts its motion from rest at a point x_0. Its mechanical energy is then $E = U(x_0)$, and this energy is conserved throughout the motion, so the starting point x_0 decides what happens to the body (Figure 8.4).

- $x_0 = \pm 1$: these are the local minimum and maximum of U, where the slope, and with it the force, is zero. If the body starts at rest at one of these points, it stays there. (But the two points behave very differently if the body is displaced slightly, as we will see in Section 8.3.1.)
- x_0 in the interval $(-1, 2)$, other than $x_0 = 1$: here $E = U(x_0) < 2m$, and the body is trapped in a *potential well*. It oscillates back and forth between two turning points at which $U = U(x_0)$, one of which is x_0 itself. It cannot climb further to the right, because it does not have enough energy to do so. Far to the left, beyond the hill at $x = -1$, there is a region where U is again smaller than E, so the conservation of energy alone would allow the body to be there. But to get there it would have to pass over the hill at $x = -1$, where $U = 2m > E$, and it does not have the energy to do so.[2]
- $x_0 > 2$: now $E = U(x_0) > 2m$. The force at x_0 points to the left, so the body speeds up toward the bottom of the well, passes through $x = 1$, where its speed has a local maximum (it does not stop there), slows down as it climbs toward $x = -1$, and goes

[2] In quantum mechanics a particle *can* sometimes get through such a barrier, a process called *tunneling*; the alpha decay of nuclei is an example. In classical mechanics it is impossible.

over the hill with a speed that is not zero, since $E > 2m$. Beyond the hill it moves toward $x \to -\infty$, faster and faster, and never comes back.

- $x_0 < -1$: the slope of U is positive, so the force points to the left, and the body moves toward $x \to -\infty$.
- $x_0 = +2$: this is a special case, in which the energy of the body is exactly the value of U at the local maximum, $E = 2m$. Released from here, the body cannot stop anywhere before $x = -1$, since $U < 2m$ everywhere between -1 and 2, so its kinetic energy is not zero there. On the other hand, at $x = -1$ both its speed and the force on it are zero, so if it ever got there, it would stay there. In fact it approaches $x = -1$ more and more slowly and gets there only in the limit $t \to \infty$. If we start the body a little to the left of $x_0 = 2$, so that E is slightly less than $2m$, it oscillates in the well, and the period of its oscillation grows without bound as E approaches $2m$; for $E = 2m$ the body never completes even one trip.

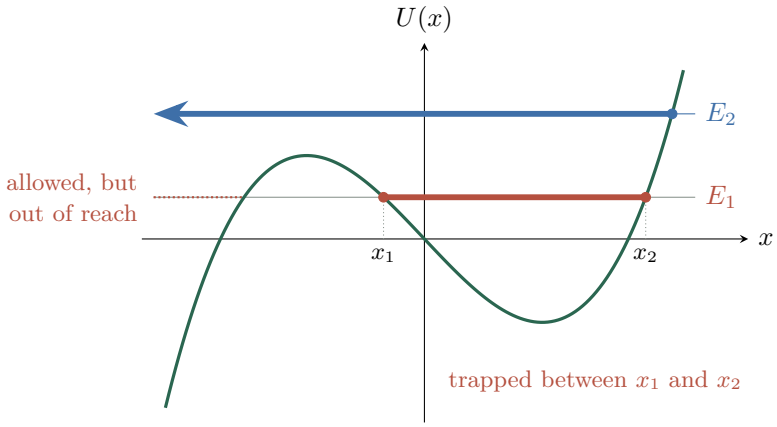


Figure 8.4. Reading an energy diagram. The body can only be where $U(x) \le E$. With the energy E_1 (red), a body in the well between -1 and 2 oscillates between the turning points x_1 and x_2; the region on the far left, where $U < E_1$ again, cannot be reached, because the hill at $x = -1$ is in the way. With the energy $E_2 > 2m$ (blue), a body released at rest at its turning point passes over the hill and moves off toward $x \to -\infty$.

We can check the last case with the integral (8.16). For $E = 2m$,

$$E - U(x) = m(2 - x^3 + 3x) = m(2 - x)(x + 1)^2,$$

and the body moves to the left, so, with $t_0 = 0$ at $x_0 = 2$,

$$t = \int_x^2 \frac{\mathrm{d}x'}{(x' + 1)\sqrt{2(2 - x')}} = \frac{1}{\sqrt{6}} \ln \frac{\sqrt{3} + \sqrt{2 - x}}{\sqrt{3} - \sqrt{2 - x}}$$

(substitute $u = \sqrt{2 - x'}$). As $x \to -1$, $\sqrt{2 - x} \to \sqrt{3}$, the denominator goes to zero, and $t \to \infty$. Near $x = -1$ the speed is about $\sqrt{6}\,(x + 1)$: it is proportional to the distance that is left, so the body covers each further stretch more slowly and never quite arrives.

8.3.1 Equilibrium

A body is in **equilibrium** at a point where the net force on it is zero: if it is placed there at rest, it stays there. In the cubic example above there are two equilibrium points,

$x = \pm 1$. In general, since $F = -\mathrm{d}U/\mathrm{d}x$, the equilibrium points are the critical points of the potential energy, where

$$\frac{\mathrm{d}U}{\mathrm{d}x} = 0. \tag{8.18}$$

But what happens if the body is not exactly at an equilibrium point, only very close to it? To see this, we call the equilibrium point x_0 (in this section x_0 no longer means the starting point) and expand the potential energy in a Taylor series around x_0:

$$U(x) = U(x_0) + \left.\frac{\mathrm{d}U}{\mathrm{d}x}\right|_{x_0} (x - x_0) + \frac{1}{2} \left.\frac{\mathrm{d}^2U}{\mathrm{d}x^2}\right|_{x_0} (x - x_0)^2 + \cdots . \tag{8.19}$$

The second term is zero, by the definition (8.18) of an equilibrium point. Close to x_0, where $x - x_0$ is small, the terms that are not written are much smaller still, and if we differentiate (8.19) we find the force

$$F = -\frac{\mathrm{d}U}{\mathrm{d}x} \approx -U''(x_0)\,(x - x_0), \qquad U''(x_0) \equiv \left.\frac{\mathrm{d}^2U}{\mathrm{d}x^2}\right|_{x_0} .$$

What happens next depends on the sign of the second derivative of the potential energy at x_0.

- $U''(x_0) > 0$: the equilibrium point is a minimum of the potential energy. Near it, the equation of motion is

  $$m\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -U''(x_0)\,(x - x_0). \tag{8.20}$$

  If we displace the body slightly from the equilibrium point, the force, and with it the acceleration, points back toward x_0, and tries to bring the body back. Equation (8.20) is the equation of a body on a spring with spring constant $k = U''(x_0)$, so the body oscillates about the minimum in simple harmonic motion, with $\omega = \sqrt{U''(x_0)/m}$. This is called **stable equilibrium**. Since near its minimum almost any smooth U looks like a parabola, small oscillations about a stable equilibrium are almost always simple harmonic.

- $U''(x_0) < 0$: the equilibrium point is a maximum of the potential energy. Near it, the equation of motion is

  $$m\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -U''(x_0)\,(x - x_0) = +\left|U''(x_0)\right|\,(x - x_0). \tag{8.21}$$

  Now if we displace the body slightly from the equilibrium point, the force points away from x_0 and pushes the body further away. The displacement grows exponentially, like $e^{\gamma t}$ with $\gamma = \sqrt{|U''(x_0)|/m}$, and the body quickly moves away from the equilibrium point. This is called **unstable equilibrium**.

If $U''(x_0) = 0$, we must look at the higher terms of the series. If U is constant near x_0, like the potential energy of a ball on a level table, the body stays wherever we put it: this is called **neutral equilibrium**. The three cases are shown in Figure 8.5.

For the cubic potential energy (8.17), $U''(x) = 6mx$. At $x = +1$, $U'' = 6m > 0$: this is a stable equilibrium, the bottom of the well, and a body slightly displaced from it oscillates with $\omega = \sqrt{6m/m} = 2.4\,\mathrm{s}^{-1}$, that is, with a period of $2\pi/\sqrt{6} \approx 2.6\,\mathrm{s}$. At $x = -1$,



(a) stable (b) unstable (c) neutral

Figure 8.5. A ball on a curved surface: its gravitational potential energy is proportional to the height of the surface. (a) At the bottom of a valley, a minimum of U, the ball returns when it is displaced. (b) On the top of a hill, a maximum of U, it rolls away. (c) On a level surface it stays wherever it is put.

$U'' = -6m < 0$: this is an unstable equilibrium, the top of the hill. (Compare the speed $\sqrt{6}\,(x+1)$ with which the body of the last case above creeps up to this point: the same $\sqrt{6} = \sqrt{|U''(-1)|/m}$ appears.)

8.4 Force and potential energy in three dimensions

It is easy to extend the ideas of conservative force and potential energy from one dimension to three. In three dimensions the potential energy is a function of all three coordinates, $U(x, y, z)$. During a small displacement $\mathrm{d}\vec{r} = \mathrm{d}x\,\hat{\imath} + \mathrm{d}y\,\hat{\jmath} + \mathrm{d}z\,\hat{k}$, a conservative force does the work

$$\mathrm{d}W = \vec{F} \cdot \mathrm{d}\vec{r} = F_x\,\mathrm{d}x + F_y\,\mathrm{d}y + F_z\,\mathrm{d}z,$$

and this must be equal to $-\mathrm{d}U$, the change in U with the opposite sign. For a small displacement, the change in U is the sum of the changes caused by $\mathrm{d}x$, $\mathrm{d}y$ and $\mathrm{d}z$ separately, each equal to the rate at which U changes in that direction times the step:

$$-\mathrm{d}U = -\frac{\partial U}{\partial x}\,\mathrm{d}x - \frac{\partial U}{\partial y}\,\mathrm{d}y - \frac{\partial U}{\partial z}\,\mathrm{d}z.$$

Here $\partial U/\partial x$ is a **partial derivative**: the derivative of U with respect to x when y and z are kept fixed, and similarly for y and z. The two expressions must agree for every small displacement; taking one along x ($\mathrm{d}y = \mathrm{d}z = 0$), then one along y, then one along z, we find

$$F_x = -\frac{\partial U}{\partial x}, \qquad F_y = -\frac{\partial U}{\partial y}, \qquad F_z = -\frac{\partial U}{\partial z}. \tag{8.22}$$

Each component of the force is minus the rate at which U changes in that direction. In vector notation we use the **gradient** operator,

$$\vec{\nabla} = \hat{\imath}\,\frac{\partial}{\partial x} + \hat{\jmath}\,\frac{\partial}{\partial y} + \hat{k}\,\frac{\partial}{\partial z}, \tag{8.23}$$

and write the force as

$$\vec{F} = -\vec{\nabla}U. \tag{8.24}$$

The gradient $\vec{\nabla}U$ points in the direction in which U increases fastest, and it is perpendicular to the surfaces on which U is constant. Think of a hiking map with contour lines, the lines along which the height of the ground is constant, drawn for equally spaced heights. The gradient of the height of the ground is a horizontal vector on the map: it points straight uphill, perpendicular to the contour lines, and it is largest where the lines are

closest together, where the slope is steepest. In the same way, $-\vec{\nabla}U$ points "downhill" in U, perpendicular to the surfaces of constant U, and the force is strongest where these surfaces, drawn for equally spaced values of U, are crowded together. For gravity near the Earth, $U = mgy$, the surfaces of constant U are horizontal planes, and $\vec{F} = -\vec{\nabla}(mgy) = -mg\,\hat{\jmath}$ points straight down, toward lower U. For a body pulled toward the origin by a force proportional to its distance from it (an idealized spring whose relaxed length is zero; this is called a three-dimensional harmonic oscillator), $U = \frac{1}{2}k(x^2 + y^2 + z^2)$ and (8.24) gives $\vec{F} = -k(x\,\hat{\imath} + y\,\hat{\jmath} + z\,\hat{k}) = -k\vec{r}$, a force that always points back to the origin.

If a force can be written in the form $\vec{F} = -\vec{\nabla}U$, it is conservative. Indeed, along any path

$$\vec{F} \cdot \mathrm{d}\vec{r} = -\Big(\frac{\partial U}{\partial x}\,\mathrm{d}x + \frac{\partial U}{\partial y}\,\mathrm{d}y + \frac{\partial U}{\partial z}\,\mathrm{d}z\Big) = -\mathrm{d}U,$$

so its work from A to B is $\int_A^B \vec{F} \cdot \mathrm{d}\vec{r} = -\int_A^B \mathrm{d}U = U_A - U_B$, which depends only on the end points. How can we tell whether a given force has this form? We use the **curl** of the force, $\vec{\nabla} \times \vec{F}$, the vector product of the operator $\vec{\nabla}$ with $\vec{F}$:

$$\vec{\nabla} \times \vec{F} = \Big(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z}\Big)\hat{\imath} + \Big(\frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x}\Big)\hat{\jmath} + \Big(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y}\Big)\hat{k}.$$

For a conservative force the curl vanishes. Take the z component, for example: with (8.22),

$$\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} = -\frac{\partial^2 U}{\partial x\,\partial y} + \frac{\partial^2 U}{\partial y\,\partial x} = 0,$$

because the order in which we take two partial derivatives does not matter; the other two components vanish in the same way. So if a force is conservative,

$$\vec{\nabla} \times \vec{F} = \vec{0}. \tag{8.25}$$

(A way to remember this: the vector product of a vector with itself, or with any vector parallel to it, is zero, and "$\vec{\nabla} \times \vec{\nabla}U$" looks like such a product. But $\vec{\nabla}$ is not an ordinary vector, and the real reason is the equality of the mixed partial derivatives.) The converse is also true for a force defined everywhere in a region with no hole running through it, like the hole of a doughnut (more precisely, a region in which every closed path can be shrunk to a point without leaving the region; all of space is such a region): if $\vec{\nabla} \times \vec{F} = \vec{0}$ there, then $\vec{F}$ is conservative. This gives us a practical test.

Example 8.4: Which force is conservative?

Consider two forces in the xy plane, $\vec{F}_1 = c(y\,\hat{\imath} + x\,\hat{\jmath})$ and $\vec{F}_2 = c(-y\,\hat{\imath} + x\,\hat{\jmath})$, where c is a positive constant. Only the z component of the curl can be nonzero. For $\vec{F}_1$ it is $\partial(cx)/\partial x - \partial(cy)/\partial y = c - c = 0$, so $\vec{F}_1$ is conservative; indeed, $U = -cxy$ gives $-\partial U/\partial x = cy$ and $-\partial U/\partial y = cx$. For $\vec{F}_2$ it is $c - (-c) = 2c \neq 0$, so $\vec{F}_2$ is not conservative. We can check this directly. On a circle of radius R around the origin, $\vec{F}_2$ is tangent to the circle, pointing counterclockwise, with magnitude cR. If a body goes once around the circle counterclockwise, the force always points along its motion and does the work $W = (cR)(2\pi R) = 2\pi cR^2$, not zero, although the body returns to where it started.

8.5 Nonconservative forces

The tools we have developed in this chapter are very powerful, but we must use them with care. The conservation of mechanical energy holds only when all the forces that do work are conservative, and not all forces are conservative. Friction, for example, is not a conservative force. To see this, consider a box that we push horizontally from a point A to a point B on a rough, horizontal floor, once along a straight line and once along a curve (Figure 8.6). The kinetic friction force has the constant magnitude $f_\mathrm{k} = \mu_\mathrm{k} n = \mu_\mathrm{k} mg$ and always points opposite to the velocity, so on every small piece $\mathrm{d}s$ of the path it does the work $-f_\mathrm{k}\,\mathrm{d}s$. Its total work is therefore $-f_\mathrm{k}$ times the *length* of the path: $-f_\mathrm{k}d$ along the straight line of length d, and $-f_\mathrm{k}s$ along the curve of length $s > d$. The work done by friction depends on the path. Around a closed path it is not zero either: if we push the box from A to B and back, friction does negative work on both legs. So friction is not a conservative force: there is no potential-energy function U whose gradient gives the friction force, $\vec{f}_\mathrm{k} = -\vec{\nabla}U$, and no potential energy can be defined for it. (Kinetic friction depends on the direction of the velocity, not only on the position of the body.) The same is true of air resistance and of the other forces that, like friction, always oppose the motion of the body.

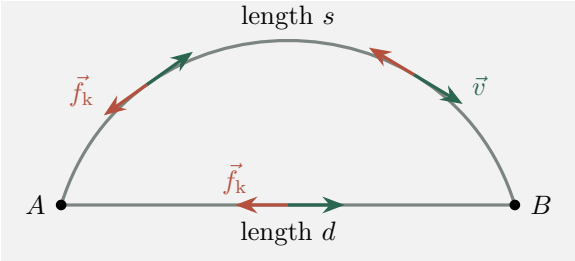


Figure 8.6. A box is pushed from A to B on a rough horizontal floor (seen from above), along a straight line of length d or along a curve of length $s > d$. Kinetic friction always points opposite to the velocity, so its work is $-f_\mathrm{k}d$ along the line and $-f_\mathrm{k}s$ along the curve: it depends on the path.

8.5.1 The work done by nonconservative forces

Does this mean that we cannot use energy at all when there is friction? Not at all. The work–energy theorem holds for the net work of *all* the forces. We split it into the work of the conservative forces, which is $-\Delta U$, and the work W_other of all the other forces: $W_\mathrm{net} = -\Delta U + W_\mathrm{other} = \Delta K$. Rearranging,

$$K_A + U_A + W_\mathrm{other} = K_B + U_B, \qquad \text{that is,} \qquad \Delta E = W_\mathrm{other}.$$

The mechanical energy changes by exactly the work done by the nonconservative forces. When we push a body, our push can increase E; the kinetic friction of a fixed surface on a body that slides over it does negative work and decreases E.

Where does the mechanical energy go? It is not destroyed: it turns into **internal energy**. Rub your hands together and they get warm; the brakes of a car get hot when it stops. Kinetic friction turns the ordered motion of the body as a whole into the random motion of

the atoms of the two surfaces, and the increase of their internal energy is $\Delta U_{\text{int}} = -W_{\text{fric}}$. If kinetic friction is the only nonconservative force that does work (nobody pushes or pulls the body), then $W_{\text{other}} = W_{\text{fric}} = -\Delta U_{\text{int}}$, and with this term included the energy is conserved after all:

$$\Delta K + \Delta U + \Delta U_{\text{int}} = 0.$$

(If we also push the body, the right-hand side is the work of our push: energy passes from us to the body.) This is the **law of conservation of energy**: energy is never created or destroyed; it only changes from one form to another, or passes from one body to another. No exception to it has ever been found. Mechanical energy, on the other hand, is conserved only when the nonconservative forces do no work.

Example 8.5: A box on a curved ramp

A box of mass 10 kg starts from rest at the top of a curved ramp, 2.0 m high, and slides down. At the bottom its speed is 5.0 m/s. How much work did friction do on it?

On a curved ramp the normal force, and so the friction force, changes along the way, and Newton's second law would be hard to use. With energy it is easy. We put $y = 0$ at the bottom. The normal force does no work, so $W_{\text{other}} = W_{\text{fric}}$, and

$$\begin{aligned} W_{\text{fric}} &= (K_B + U_B) - (K_A + U_A) = \tfrac{1}{2}(10\,\text{kg})(5.0\,\text{m/s})^2 - (10\,\text{kg})(9.8\,\text{m/s}^2)(2.0\,\text{m}) \\ &= 125\,\text{J} - 196\,\text{J} = -71\,\text{J}. \end{aligned}$$

Of the 196 J of potential energy that the box had at the top, 125 J became kinetic energy and 71 J became internal energy of the box and the ramp, which got slightly warmer.

About these notes

These notes are not original. They draw heavily on the books listed in the Week 1 notes, on Walter Lewin's 8.01x Classical Mechanics lectures at MIT, and on the Dynamics and Relativity courses taught at different times by Stephen Siklos and David Tong at the University of Cambridge. They are not official course notes of FIZ 101: they are written only to help the students of Mehmet Özkan's section. Istanbul Technical University and its Department of Physics Engineering bear no responsibility for them.